

Algebraic Geometry 18

Riemann surfaces

Contents

1 Problem 1: Winding Numbers in a Slit Plane

Statement

Let

$$U = \mathbb{C} \setminus ([-1, 0] \cup [1, \infty))$$

and let γ be a closed curve in U .

Using standard properties of winding numbers, show that

1.

$$n(\gamma, 1) = 0,$$

2.

$$n(\gamma, 0) = n(\gamma, -1).$$

Solution

Recall that the winding number of γ about a point $a \notin \gamma$ is

$$n(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - a}.$$

We use the following standard facts.

- If a and b lie in the same connected component of

$$\mathbb{C} \setminus \gamma,$$

then

$$n(\gamma, a) = n(\gamma, b).$$

- If a belongs to the unbounded component of

$$\mathbb{C} \setminus \gamma,$$

then

$$n(\gamma, a) = 0.$$

Part (i)

Since

$$[1, \infty) \subset \mathbb{C} \setminus U,$$

the curve γ cannot cross this ray.

Consequently there exists a path from 1 to infinity entirely contained in

$$\mathbb{C} \setminus \gamma.$$

Therefore 1 lies in the same connected component of

$$\mathbb{C} \setminus \gamma$$

as infinity.

Hence, by the second property,

$$n(\gamma, 1) = 0.$$

Thus

$$\boxed{n(\gamma, 1) = 0.}$$

Part (ii)

The segment

$$[-1, 0]$$

lies entirely in the complement of U .

Since $\gamma \subset U$, the curve does not intersect this segment.

Hence 0 and -1 can be connected by a path that avoids γ .

Therefore 0 and -1 belong to the same connected component of

$$\mathbb{C} \setminus \gamma.$$

Applying the first property gives

$$n(\gamma, 0) = n(\gamma, -1).$$

Therefore

$$\boxed{n(\gamma, 0) = n(\gamma, -1).}$$

Combining both parts,

$$\boxed{n(\gamma, 1) = 0, \quad n(\gamma, 0) = n(\gamma, -1).}$$

2 Problem 2: Analytic Continuation of Solutions of Differential Equations

Statement

Let

$$P(w_0, w_1, \dots, w_s; z)$$

be a polynomial in the variables $w_0, \dots, w_s$, whose coefficients are holomorphic functions of $z \in \mathbb{C}$.

Then

$$P(f(z), f'(z), \dots, f^{(s)}(z); z) = 0$$

is a differential equation, abbreviated as

$$P(f) = 0.$$

Suppose (f, D) is a function element satisfying $P(f) = 0$ in D , and

$$(g, D') \approx (f, D)$$

is an analytic continuation.

Show that

$$P(g) = 0$$

in D' .

Give an example of a differential equation $P(f) = 0$ and function elements

$$(f, D) \approx (g, D')$$

such that

$$P(f) = P(g) = 0$$

but

$$g \neq f.$$

Solution

Let

$$(g, D') \approx (f, D).$$

By definition of analytic continuation there exists a chain of overlapping domains through which f continues analytically to g .

On each overlap,

$$f = g.$$

Since equal holomorphic functions have equal derivatives,

$$f^{(k)} = g^{(k)}$$

for every k .

Substituting into the polynomial expression gives

$$P(f) = P(g)$$

on the overlap.

Because $P(f) = 0$,

$$P(g) = 0$$

on a nonempty open set.

Now $P(g)$ is itself a holomorphic function.

By the Identity Theorem,

$$P(g) \equiv 0$$

throughout the connected continuation domain.

Therefore

$$\boxed{P(g) = 0.}$$

Thus every analytic continuation of a solution remains a solution of the differential equation.

Example

Consider the differential equation

$$P(f) = f' - \frac{1}{2f} = 0.$$

This is equivalent to

$$2ff' = 1.$$

Since

$$(f^2)' = 2ff',$$

we obtain

$$(f^2)' = 1.$$

Integrating,

$$f^2 = z + C.$$

Take the branch solutions

$$f(z) = \sqrt{z}, \quad g(z) = -\sqrt{z}.$$

Then

$$f^2 = g^2 = z.$$

Furthermore,

$$f' = \frac{1}{2f}, \quad g' = \frac{1}{2g},$$

so both satisfy

$$P(f) = 0, \quad P(g) = 0.$$

However,

$$g = -f,$$

and therefore

$$g \neq f.$$

Hence

$$P(f) = P(g) = 0, \quad g \neq f.$$

3 Problem 3: Fixed Points of Deck Transformations

Statement

Let

$$\pi : \widetilde{X} \rightarrow X$$

be a covering map of path-connected Hausdorff spaces.

Suppose

$$f : \widetilde{X} \rightarrow \widetilde{X}$$

is continuous and satisfies

$$\pi \circ f = \pi.$$

Show that f has no fixed points unless

$$f = \text{id}_{\widetilde{X}}.$$

Solution

The condition

$$\pi \circ f = \pi$$

means that f preserves every fiber of the covering.

Hence f is a deck transformation (covering transformation).

Assume that f has a fixed point

$$\tilde{x}_0 \in \widetilde{X}.$$

Then

$$f(\tilde{x}_0) = \tilde{x}_0.$$

Consider the identity map

$$\mathrm{id}_{\widetilde{X}} : \widetilde{X} \rightarrow \widetilde{X}.$$

We have

$$\pi \circ f = \pi = \pi \circ \mathrm{id}_{\widetilde{X}}.$$

Thus f and $\mathrm{id}_{\widetilde{X}}$ are two lifts of the same map π .

Moreover,

$$f(\widetilde{x}_0) = \mathrm{id}_{\widetilde{X}}(\widetilde{x}_0).$$

By the uniqueness theorem for lifts, two lifts of the same map that agree at one point must agree everywhere.

Therefore

$$f = \mathrm{id}_{\widetilde{X}}.$$

We conclude that if $f \neq \mathrm{id}_{\widetilde{X}}$, then f cannot possess any fixed point.

Hence

$$\boxed{f \text{ has a fixed point} \implies f = \mathrm{id}_{\widetilde{X}}.}$$

Equivalently,

$$\boxed{\text{Every nontrivial deck transformation acts freely.}}$$

4 Problem 4: Analytic Continuation of a Power Series

Statement

Show that the power series

$$f(z) = \sum_{n=2}^{\infty} \frac{1}{n(n-1)} z^n$$

converges on the unit disc

$$D = D(0, 1)$$

to the analytic function

$$z + (1 - z) \log(1 - z).$$

Deduce that the function element (f, D) can be analytically continued to every point of

$$\mathbb{C} \setminus \{1\}$$

as a complete analytic function, but that it does not extend to an analytic function on

$$\mathbb{C} \setminus \{1\}.$$

Solution

Observe that

$$\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}.$$

Hence

$$f(z) = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) z^n.$$

Splitting the series gives

$$f(z) = \sum_{n=2}^{\infty} \frac{z^n}{n-1} - \sum_{n=2}^{\infty} \frac{z^n}{n}.$$

For the first sum, letting $m = n - 1$,

$$\sum_{n=2}^{\infty} \frac{z^n}{n-1} = z \sum_{m=1}^{\infty} \frac{z^m}{m}.$$

Using

$$-\log(1-z) = \sum_{m=1}^{\infty} \frac{z^m}{m}, \quad |z| < 1,$$

we obtain

$$\sum_{n=2}^{\infty} \frac{z^n}{n-1} = -z \log(1-z).$$

Similarly,

$$\sum_{n=2}^{\infty} \frac{z^n}{n} = \sum_{n=1}^{\infty} \frac{z^n}{n} - z = -\log(1-z) - z.$$

Therefore

$$f(z) = -z \log(1 - z) + \log(1 - z) + z.$$

Factoring gives

$$f(z) = z + (1 - z) \log(1 - z).$$

Thus

$$f(z) = z + (1 - z) \log(1 - z).$$

Analytic Continuation

The logarithm

$$\log(1 - z)$$

can be analytically continued along every path avoiding the point 1.

Consequently the function element (f, D) can be analytically continued to every point of

$$\mathbb{C} \setminus \{1\}.$$

However, analytic continuation around a loop encircling 1 changes the logarithm by

$$2\pi i.$$

Hence

$$z + (1 - z) \log(1 - z)$$

changes by

$$2\pi i(1 - z).$$

Since this is not identically zero, the continuation is multivalued.

Therefore no single-valued holomorphic function exists on all of

$$\mathbb{C} \setminus \{1\}$$

whose restriction to D equals the original power series.

Hence

$$(f, D) \text{ can be analytically continued throughout } \mathbb{C} \setminus \{1\},$$

but

f does not extend to a single-valued analytic function on $\mathbb{C} \setminus \{1\}$.

5 Problem 5: A Natural Boundary

Statement

Show that the power series

$$f(z) = \sum_{n=0}^{\infty} \frac{z^{2^n}}{2^n}$$

has the unit circle as a natural boundary.

Solution

The exponents occurring in the series are

$$1, 2, 4, 8, 16, \dots, 2^n, \dots$$

and therefore increase exponentially.

The coefficient of z^{2^n} is

$$a_{2^n} = \frac{1}{2^n}.$$

The radius of convergence is determined by

$$R^{-1} = \limsup_{m \rightarrow \infty} |a_m|^{1/m}.$$

Since

$$\left(\frac{1}{2^n}\right)^{1/2^n} \longrightarrow 1,$$

we obtain

$$R = 1.$$

Thus the series converges precisely in

$$|z| < 1.$$

Hadamard Gap Theorem

The exponents satisfy

$$m_n = 2^n, \quad \frac{m_{n+1}}{m_n} = 2.$$

Hence

$$\frac{m_{n+1}}{m_n} \geq q$$

with

$$q = 2 > 1.$$

Therefore the series is a lacunary (gap) series.

The Hadamard Gap Theorem states that if

$$\frac{m_{n+1}}{m_n} \geq q > 1,$$

then every point of the circle of convergence is a singular point.

Consequently every point of

$$|z| = 1$$

is singular.

Thus no analytic continuation is possible across any arc of the unit circle.

Hence

$$|z| = 1 \text{ is a natural boundary of } f.$$

Equivalently,

$$f \text{ cannot be analytically continued through any point of } |z| = 1.$$

6 Problem 6: Equivalence of Atlases and Maximal Atlases

Statement

Show that equivalence of atlases is an equivalence relation.

Show that every conformal structure contains a unique maximal atlas.

Solution

Let X be a topological surface.

A conformal atlas consists of charts

$$(U_\alpha, \varphi_\alpha)$$

such that

$$\varphi_\alpha : U_\alpha \rightarrow V_\alpha \subset \mathbb{C}$$

is a homeomorphism and every transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1}$$

is holomorphic with holomorphic inverse.

Two atlases $\mathcal{A}$ and $\mathcal{B}$ are called equivalent if

$$\mathcal{A} \cup \mathcal{B}$$

is again a conformal atlas.

Reflexivity

For any atlas $\mathcal{A}$,

$$\mathcal{A} \cup \mathcal{A} = \mathcal{A}.$$

Hence

$$\mathcal{A} \sim \mathcal{A}.$$

Symmetry

If

$$\mathcal{A} \sim \mathcal{B},$$

then

$$\mathcal{A} \cup \mathcal{B}$$

is conformal.

Since

$$\mathcal{A} \cup \mathcal{B} = \mathcal{B} \cup \mathcal{A},$$

we obtain

$$\mathcal{B} \sim \mathcal{A}.$$

Transitivity

Assume

$$\mathcal{A} \sim \mathcal{B}, \quad \mathcal{B} \sim \mathcal{C}.$$

Take charts

$$(U, \varphi) \in \mathcal{A}, \quad (W, \psi) \in \mathcal{C}.$$

Since the charts of $\mathcal{B}$ cover the surface, every point of

$$U \cap W$$

lies in some chart

$$(V, \theta) \in \mathcal{B}.$$

On the overlap,

$$\psi \circ \varphi^{-1} = (\psi \circ \theta^{-1}) \circ (\theta \circ \varphi^{-1}).$$

Both factors are conformal because of compatibility with $\mathcal{B}$.

Therefore their composition is conformal.

Hence every chart of $\mathcal{A}$ is compatible with every chart of $\mathcal{C}$.

Thus

$$\mathcal{A} \sim \mathcal{C}.$$

Therefore atlas equivalence is reflexive, symmetric, and transitive.

Hence

equivalence of atlases is an equivalence relation.

Maximal Atlas

Let $\mathcal{A}$ be a conformal atlas.

Define

$$\mathcal{A}_{\max} = \{(U, \varphi) : (U, \varphi) \text{ is compatible with } \mathcal{A}\}.$$

Every chart in $\mathcal{A}_{\max}$ is compatible with every chart of $\mathcal{A}$.

Consequently any two charts of $\mathcal{A}_{\max}$ are compatible with one another.

Therefore

$$\mathcal{A}_{\max}$$

is itself a conformal atlas.

Clearly,

$$\mathcal{A} \subseteq \mathcal{A}_{\max}.$$

Now let $\mathcal{B}$ be any atlas equivalent to $\mathcal{A}$.

Every chart of $\mathcal{B}$ is compatible with $\mathcal{A}$.

Hence

$$\mathcal{B} \subseteq \mathcal{A}_{\max}.$$

Therefore $\mathcal{A}_{\max}$ contains every atlas belonging to the same conformal structure.

Thus $\mathcal{A}_{\max}$ is maximal.

Uniqueness

Suppose

$$\mathcal{M}$$

is another maximal atlas in the same conformal structure.

Since every chart of $\mathcal{M}$ is compatible with $\mathcal{A}$,

$$\mathcal{M} \subseteq \mathcal{A}_{\max}.$$

By maximality of $\mathcal{M}$,

$$\mathcal{A}_{\max} \subseteq \mathcal{M}.$$

Therefore

$$\mathcal{M} = \mathcal{A}_{\max}.$$

Hence the maximal atlas is unique.

Thus

every conformal structure contains a unique maximal atlas.

7 Problem 7: Transition Functions on a Complex Torus

Statement

Let

$$T = \mathbb{C}/\langle 1, \tau \rangle$$

be the complex torus, and let $Q_1 \subset \mathbb{C}$ be the open parallelogram with vertices

$$0, \quad 1, \quad \tau, \quad 1 + \tau,$$

and Q_2 the translate of Q_1 by

$$\frac{1 + \tau}{2}.$$

Let

$$U_1, U_2 \subset T$$

denote the images of Q_1, Q_2 under the quotient projection

$$p : \mathbb{C} \rightarrow T.$$

Let

$$\phi_1 : U_1 \rightarrow Q_1, \quad \phi_2 : U_2 \rightarrow Q_2$$

be the corresponding inverse charts.

Describe explicitly the transition function

$$\phi_2 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2) \rightarrow \phi_2(U_1 \cap U_2).$$

Solution

Write

$$z = x + y\tau, \quad 0 < x < 1, \quad 0 < y < 1.$$

Then

$$Q_1 = \{x + y\tau : 0 < x < 1, \ 0 < y < 1\}.$$

Since

$$Q_2 = Q_1 + \frac{1 + \tau}{2},$$

we have

$$Q_2 = \left\{ x + y\tau : \frac{1}{2} < x < \frac{3}{2}, \ \frac{1}{2} < y < \frac{3}{2} \right\}.$$

For a point $z \in Q_1$, the transition map must choose the unique representative of the same lattice class that lies inside Q_2 .

Thus one adds suitable lattice vectors from

$$\Lambda = \langle 1, \tau \rangle.$$

Define

$$\varepsilon_1 = \begin{cases} 1, & 0 < x < \frac{1}{2}, \\ 0, & \frac{1}{2} < x < 1, \end{cases}$$

and

$$\varepsilon_2 = \begin{cases} 1, & 0 < y < \frac{1}{2}, \\ 0, & \frac{1}{2} < y < 1. \end{cases}$$

Then

$$\phi_2 \circ \phi_1^{-1}(z) = z + \varepsilon_1 + \varepsilon_2\tau.$$

Hence

$$\phi_2 \circ \phi_1^{-1}(z) = \begin{cases} z + 1 + \tau, & 0 < x < \frac{1}{2}, \quad 0 < y < \frac{1}{2}, \\ z + 1, & 0 < x < \frac{1}{2}, \quad \frac{1}{2} < y < 1, \\ z + \tau, & \frac{1}{2} < x < 1, \quad 0 < y < \frac{1}{2}, \\ z, & \frac{1}{2} < x < 1, \quad \frac{1}{2} < y < 1. \end{cases}$$

Each branch is a translation by a lattice element and therefore holomorphic.

Consequently the transition map is conformal.

$\phi_2 \circ \phi_1^{-1}$ is piecewise a lattice translation.

8 Problem 8: Injective Entire Functions

Statement

By considering the singularity at ∞ , show that every injective analytic map

$$f : \mathbb{C} \rightarrow \mathbb{C}$$

has the form

$$f(z) = az + b, \quad a \in \mathbb{C}^*, \quad b \in \mathbb{C}.$$

Characterise the injective analytic maps

$$\mathbb{C}_\infty \rightarrow \mathbb{C}_\infty.$$

Solution

Let

$$f : \mathbb{C} \rightarrow \mathbb{C}$$

be injective and entire.

Consider the behaviour of f at infinity.

Introduce

$$g(w) = f\left(\frac{1}{w}\right).$$

The point $w = 0$ corresponds to $z = \infty$.

There are three possibilities.

Case 1: Infinity is removable

Then f is bounded near infinity.

Since f is also bounded on every compact disc, it is bounded on all of $\mathbb{C}$.

By Liouville's theorem,

$$f$$

must be constant.

This contradicts injectivity.

Therefore infinity cannot be removable.

Case 2: Infinity is essential

Suppose infinity is an essential singularity.

Then, by the Casorati–Weierstrass theorem, the image of every punctured neighbourhood of infinity is dense in $\mathbb{C}$.

Consequently values already attained by f inside a compact disc are approximated arbitrarily closely by values of f outside the disc.

Injectivity becomes impossible.

Hence infinity cannot be essential.

Case 3: Infinity is a pole

The only remaining possibility is that infinity is a pole.

Therefore f is a polynomial:

$$f(z) = a_n z^n + \cdots + a_1 z + a_0.$$

If $n \geq 2$, then a generic value has n preimages.

Therefore f cannot be injective.

Hence

$$n = 1.$$

Thus

$$f(z) = az + b, \quad a \neq 0.$$

We conclude that

$$f(z) = az + b, \quad a \in \mathbb{C}^*, \quad b \in \mathbb{C}.$$

Maps of the Riemann Sphere

Now consider

$$f : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty.$$

Every holomorphic self-map of the sphere is rational.

Injectivity implies bijectivity.

Hence f must be a Möbius transformation:

$$f(z) = \frac{az + b}{cz + d},$$

with

$$ad - bc \neq 0.$$

Therefore

$$\text{Aut}(\mathbb{C}_\infty) = \left\{ z \mapsto \frac{az + b}{cz + d} : ad - bc \neq 0 \right\}.$$

9 Problem 9: Conformal Equivalence of Complex Tori

Statement

Let

$$\Lambda = \langle \tau_1, \tau_2 \rangle$$

be a lattice in $\mathbb{C}$, and let

$$T = \mathbb{C}/\Lambda$$

be the corresponding complex torus.

Define

$$\Lambda' = \left\langle 1, \frac{\tau_2}{\tau_1} \right\rangle$$

and let

$$T' = \mathbb{C}/\Lambda'.$$

Show that T and T' are conformally equivalent.

Solution

Consider the linear map

$$F : \mathbb{C} \rightarrow \mathbb{C}, \quad F(z) = \frac{z}{\tau_1}.$$

Since $\tau_1 \neq 0$, this map is biholomorphic.

Furthermore,

$$F(\tau_1) = 1,$$

and

$$F(\tau_2) = \frac{\tau_2}{\tau_1}.$$

Hence

$$F(\Lambda) = \Lambda'.$$

Indeed,

$$F(m\tau_1 + n\tau_2) = m + n\frac{\tau_2}{\tau_1},$$

which belongs to Λ' .

Thus lattice-equivalent points are mapped to lattice-equivalent points.

Therefore F descends to a map

$$\tilde{F} : \mathbb{C}/\Lambda \longrightarrow \mathbb{C}/\Lambda'$$

defined by

$$\tilde{F}([z]) = \left[\frac{z}{\tau_1} \right].$$

This map is well defined and holomorphic.

Its inverse is induced by

$$w \mapsto \tau_1 w.$$

Hence $\tilde{F}$ is biholomorphic.

Therefore

$$\mathbb{C}/\langle \tau_1, \tau_2 \rangle \cong \mathbb{C}/\left\langle 1, \frac{\tau_2}{\tau_1} \right\rangle$$

as Riemann surfaces.

Consequently

$$T \text{ and } T' \text{ are conformally equivalent.}$$

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Solution

Consider the linear map

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Since $\tau_1 \neq 0$, this map is biholomorphic.

Furthermore,

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Therefore

$$\mathbb{C}/\langle \tau_1, \tau_2 \rangle \cong \mathbb{C}/\left\langle 1, \frac{\tau_2}{\tau_1} \right\rangle$$

as Riemann surfaces.

Consequently

$$T \text{ and } T' \text{ are conformally equivalent.}$$

13 Theory Behind Problems 1–9: Analytic Continuation, Coverings, and Complex Tori

The first nine problems form a remarkably coherent introduction to several central themes of complex analysis and Riemann surface theory. Their common thread is the passage from local analytic information to global geometric structures.

At a conceptual level, the material follows the progression

$$\text{Local Analytic Data} \longrightarrow \text{Analytic Continuation} \longrightarrow \text{Riemann Surfaces} \longrightarrow \text{Complex Tori}$$

and simultaneously

$$\text{Power Series} \longrightarrow \text{Singularities} \longrightarrow \text{Natural Boundaries} \longrightarrow \text{Global Geometry.}$$

13.1 Winding Numbers and Homotopy Theory

For a closed curve

$$\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{a\},$$

the winding number of γ around a is

$$n(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - a}.$$

Geometrically, $n(\gamma, a)$ counts how many times the curve winds around the point a .

Typical examples are

$$n(\gamma, a) = 0, \quad n(\gamma, a) = 1, \quad n(\gamma, a) = -1,$$

corresponding respectively to no winding, one positive winding, and one negative winding.

The fundamental property is that

$$a \mapsto n(\gamma, a)$$

is constant on every connected component of

$$\mathbb{C} \setminus \gamma.$$

Thus winding numbers are topological invariants.

This principle is precisely what underlies Problem 1.

Connection with Cauchy's Formula

Winding numbers appear naturally in Cauchy's integral formula:

$$f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz.$$

More generally,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz = n(\gamma, a) f(a).$$

Thus winding numbers are hidden inside every contour integral.

Connection with Topology

The punctured plane satisfies

$$\pi_1(\mathbb{C}^\times) = \mathbb{Z}.$$

The winding number is exactly the integer associated with a loop under this identification.

Hence

Winding Numbers $\Longleftrightarrow$ Fundamental Groups $\Longleftrightarrow$ Covering Spaces.

13.2 Differential Equations and Analytic Continuation

Suppose

$$P(f, f', \dots, f^{(s)}; z) = 0$$

is a polynomial differential equation.

The fundamental principle is:

Analytic continuation preserves algebraic and differential relations.

Indeed, if two analytic functions agree on an overlap, then all derivatives agree:

$$f = g \implies f^{(k)} = g^{(k)}.$$

Therefore any polynomial relation among derivatives continues automatically.

This observation explains why many special functions extend far beyond their initial domains.

Examples include

$$e^z, \quad \sin z, \quad \Gamma(z), \quad \wp(z).$$

Monodromy

Analytic continuation around a closed loop need not return the original branch.

For example,

$$f(z) = \sqrt{z}$$

changes according to

$$\sqrt{z} \mapsto -\sqrt{z}$$

after one turn around 0.

This phenomenon is called *monodromy*.

Problem 2 represents the first appearance of monodromy theory.

13.3 Covering Spaces and Deck Transformations

A covering map

$$\pi : \widetilde{X} \rightarrow X$$

is a local homeomorphism.

Locally, $\widetilde{X}$ looks like several copies of X .

The fundamental example is

$$\pi(z) = e^z,$$

which gives

$$\mathbb{C} \longrightarrow \mathbb{C}^\times.$$

Every point has infinitely many preimages:

$$z + 2\pi ik, \quad k \in \mathbb{Z}.$$

Deck Transformations

A deck transformation satisfies

$$\pi \circ f = \pi.$$

For the exponential covering,

$$f(z) = z + 2\pi ik.$$

The collection of all deck transformations forms a group:

$$\text{Deck}(\widetilde{X}/X).$$

Problem 3 proves the fundamental theorem:

Every nontrivial deck transformation has no fixed points.

Universal Coverings

For universal coverings one has

$$\text{Deck}(\widetilde{X}/X) \cong \pi_1(X).$$

This is one of the deepest connections between topology and geometry.

13.4 Logarithms and Branch Points

Problem 4 transforms a power series into

$$f(z) = z + (1 - z) \log(1 - z).$$

Thus everything reduces to understanding

$$\log(1 - z).$$

The special point is

$$z = 1.$$

Near this point,

$$\log(1 - z) = \ln |1 - z| + i \arg(1 - z).$$

After one circuit around 1,

$$\arg(1 - z) \mapsto \arg(1 - z) + 2\pi,$$

and hence

$$\log(1 - z) \mapsto \log(1 - z) + 2\pi i.$$

Thus $z = 1$ is not an ordinary singularity but a branch point.

Branch Points versus Poles

Poles behave like

$$\frac{1}{z}.$$

Essential singularities behave like

$$e^{1/z}.$$

Branch points behave like

$$\sqrt{z}, \quad \log z.$$

Unlike poles, branch points produce multiple sheets.

13.5 Natural Boundaries

A natural boundary is one of the most striking phenomena in complex analysis.

For many familiar functions there is only a finite set of singularities.

For example,

$$\frac{1}{1-z}$$

has a single singularity at $z = 1$.

A natural boundary is completely different.

Every boundary point is singular.

No analytic continuation is possible.

Hadamard Gap Theorem

Suppose

$$f(z) = \sum a_n z^{m_n}$$

with

$$\frac{m_{n+1}}{m_n} > q > 1.$$

Then the circle of convergence is a natural boundary.

Problem 5 uses

$$m_n = 2^n.$$

Since

$$\frac{m_{n+1}}{m_n} = 2,$$

the theorem applies directly.

Hence

$$|z| = 1$$

is a natural boundary.

13.6 Atlases and Conformal Structures

A Riemann surface is built from local charts

$$(U, \phi),$$

where

$$\phi : U \rightarrow \mathbb{C}$$

is a homeomorphism onto an open subset of $\mathbb{C}$.

The crucial requirement is that transition maps

$$\phi_j \circ \phi_i^{-1}$$

are holomorphic.

This is the fundamental definition of a complex manifold of dimension one.

Maximal Atlases

Starting with one conformal atlas, one may enlarge it by including every compatible chart.

The result is a maximal atlas.

Thus a Riemann surface is properly viewed as

$$(X, \mathcal{A}_{\max}).$$

Problem 6 proves that every conformal structure contains a unique maximal atlas.

13.7 Complex Tori

Let

$$\Lambda = \langle 1, \tau \rangle$$

be a lattice.

The corresponding complex torus is

$$T = \mathbb{C}/\Lambda.$$

Points differing by a lattice vector are identified:

$$z \sim z + m + n\tau.$$

Fundamental Parallelogram

A fundamental domain is

$$Q = \{x + y\tau : 0 \leq x, y < 1\}.$$

Opposite sides are identified.

Topologically this produces a torus.

$$\text{Parallelogram} \longrightarrow \text{Torus}.$$

Because all transition maps are translations,

$$z \mapsto z + \lambda,$$

the resulting surface is naturally a Riemann surface.

Problem 7 constructs this complex structure explicitly.

13.8 Injective Entire Functions

Problem 8 classifies injective holomorphic maps

$$\mathbb{C} \rightarrow \mathbb{C}.$$

The singularity at infinity may be

1. removable,
2. essential,
3. a pole.

The first two possibilities are impossible.

Therefore infinity must be a pole.

Hence f is a polynomial.

Injectivity forces degree one.

Thus

$$\boxed{f(z) = az + b, \quad a \neq 0.}$$

Therefore

$$\text{Aut}(\mathbb{C}) = \{az + b\}.$$

The Sphere

For the Riemann sphere,

$$\hat{\mathbb{C}} = \mathbb{C}_\infty,$$

the automorphism group is

$$\mathrm{PSL}(2, \mathbb{C}).$$

Every automorphism has the form

$$z \mapsto \frac{az + b}{cz + d}, \quad ad - bc \neq 0.$$

13.9 Moduli of Complex Tori

Let

$$\Lambda = \langle \tau_1, \tau_2 \rangle.$$

Dividing by τ_1 transforms the lattice into

$$\Lambda' = \left\langle 1, \frac{\tau_2}{\tau_1} \right\rangle.$$

Hence every torus may be written in the normalized form

$$T_\tau = \mathbb{C} / \langle 1, \tau \rangle.$$

Problem 9 proves that this normalization does not change the conformal type.

Upper Half Plane

One usually chooses

$$\tau \in \mathbb{H} = \{z \in \mathbb{C} : \Im \tau > 0\}.$$

Thus every complex torus corresponds to a point of the upper half-plane.

Modular Transformations

Different values of τ may determine the same torus.

The equivalence relation is

$$\tau \sim \frac{a\tau + b}{c\tau + d},$$

where

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z}).$$

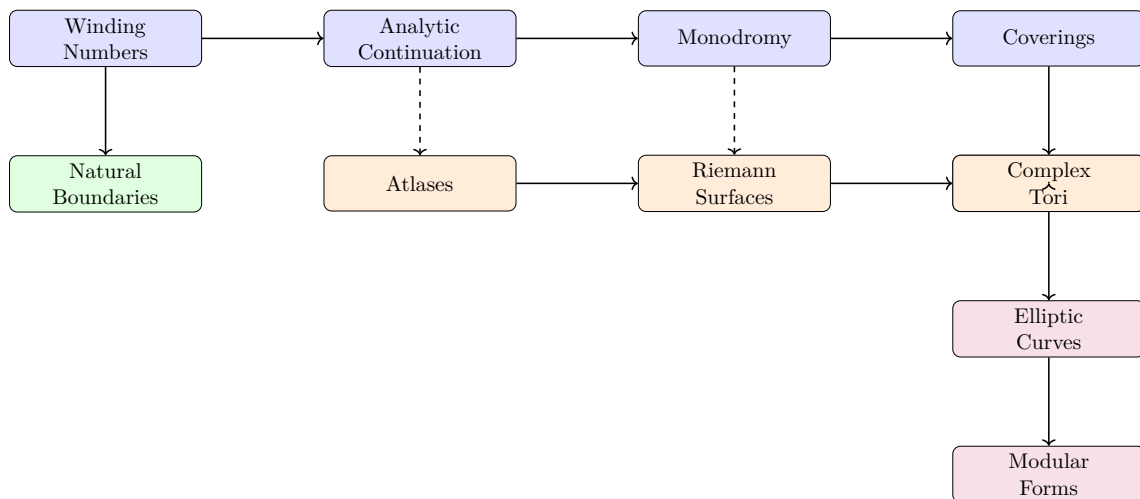
The resulting quotient

$$\mathcal{M}_1 = \mathbb{H}/SL(2, \mathbb{Z})$$

is the moduli space of elliptic curves.

13.10 Grand Synthesis

The conceptual flow of Problems 1–9 can be summarized by the following diagram.



Thus Problems 1–9 form a miniature introduction to the modern theory of

Riemann Surfaces, Covering Spaces, Elliptic Functions,
 Complex Tori, Algebraic Curves, Modular Forms.

14 Problem 10: A Quotient of $\mathbb{C}^*$ as a Complex Torus

Statement

Define an equivalence relation on

$$\mathbb{C}^* = \mathbb{C} \setminus \{0\}$$

by

$$z \sim w$$

if

$$z = 2^s w$$

for some

$$s \in \mathbb{Z}.$$

Show that the quotient space

$$R = \mathbb{C}^* / \sim$$

has the natural structure of a compact Riemann surface, and that R is analytically isomorphic to a complex torus.

Solution

The key idea is to convert multiplication into addition using the logarithm.

Step 1: Pass to the Universal Covering

Recall that

$$\exp : \mathbb{C} \rightarrow \mathbb{C}^*$$

is the universal covering map.

Write

$$z = e^u.$$

The relation

$$z \sim 2^s z$$

becomes

$$e^u \sim e^{u+s \log 2}.$$

Since

$$e^{u+2\pi i k} = e^u, \quad k \in \mathbb{Z},$$

the coordinate u is identified by

$$u \sim u + s \log 2 + 2\pi i k, \quad s, k \in \mathbb{Z}.$$

Thus the quotient can be written as

$$R \cong \mathbb{C} / \langle \log 2, 2\pi i \rangle.$$

Step 2: The Associated Lattice

Define

$$\Lambda = \langle \log 2, 2\pi i \rangle.$$

Since

$$\log 2$$

and

$$2\pi i$$

are linearly independent over $\mathbb{R}$,

$$\Lambda$$

is a lattice in $\mathbb{C}$.

Hence

$$\mathbb{C} / \Lambda$$

is a complex torus.

Therefore

$$R \cong \mathbb{C} / \Lambda.$$

Step 3: Compactness

A fundamental parallelogram for the lattice is

$$P = \{a \log 2 + b(2\pi i) : 0 \leq a, b < 1\}.$$

Every equivalence class has a representative in P .

Since P is compact, the quotient

$$\mathbb{C}/\Lambda$$

is compact.

Hence

$$R$$

is compact.

Step 4: Complex Structure

Translations

$$u \mapsto u + \lambda, \quad \lambda \in \Lambda,$$

are holomorphic maps.

Therefore the quotient inherits a natural complex structure.

Hence

$$R$$

is a compact Riemann surface.

Conclusion

$$R = \mathbb{C}^* / \{z \sim 2^s z\} \cong \mathbb{C} / \langle \log 2, 2\pi i \rangle.$$

Therefore

$$R \text{ is a compact Riemann surface and a complex torus.}$$

15 Problem 11: Identity Principle for Riemann Surfaces

Statement

Let R, S be Riemann surfaces and let

$$f, g : R \rightarrow S$$

be analytic maps.

Define

$$E = \{z \in R : f(z) = g(z)\}.$$

Show that either

$$E = R$$

or E contains only isolated points.

Solution

The proof is local and reduces the statement to the classical identity theorem for holomorphic functions.

Let

$$p \in E.$$

Then

$$f(p) = g(p).$$

Choose charts

$$(U, \phi)$$

around p and

$$(V, \psi)$$

around

$$f(p) = g(p).$$

Define local representatives

$$F = \psi \circ f \circ \phi^{-1}, \quad G = \psi \circ g \circ \phi^{-1}.$$

These are holomorphic functions on an open subset of $\mathbb{C}$.

The set E corresponds locally to

$$\{w : F(w) = G(w)\}.$$

Define

$$H(w) = F(w) - G(w).$$

Then H is holomorphic.

Case 1: $H \equiv 0$

If

$$H \equiv 0,$$

then

$$F = G$$

on the chart.

Therefore

$$f = g$$

on a neighborhood of p .

By analytic continuation and connectedness of R ,

$$f = g$$

everywhere on R .

Hence

$$E = R.$$

Case 2: $H \not\equiv 0$

If H is not identically zero, then the classical identity theorem implies that its zeros are isolated.

Consequently

$$\{H = 0\}$$

consists only of isolated points.

Thus E contains only isolated points.

Conclusion

Exactly one of the following alternatives holds:

$$E = R,$$

or

$$E \text{ consists entirely of isolated points.}$$

This is the identity principle for Riemann surfaces.

16 Problem 12: Harmonic Functions and Holomorphic Functions

Statement

Let

$$D \subset \mathbb{C}$$

be an open disc and let

$$u : D \rightarrow \mathbb{R}$$

be harmonic.

Define

$$g = u_x - iu_y.$$

Show that g is analytic.

If z_0 denotes the center of the disc, define

$$f(z) = u(z_0) + \int_{z_0}^z g(t) dt,$$

where the integral is taken along the straight line segment from z_0 to z .

Show that f is analytic with

$$f' = g,$$

and that

$$u = \operatorname{Re}(f).$$

Solution

Since u is harmonic,

$$u_{xx} + u_{yy} = 0.$$

Define

$$g = u_x - iu_y.$$

Write

$$g = P + iQ,$$

where

$$P = u_x, \quad Q = -u_y.$$

We verify the Cauchy–Riemann equations.

First Cauchy–Riemann Equation

We have

$$P_x = u_{xx}.$$

Also,

$$Q_y = -u_{yy}.$$

Since

$$u_{xx} + u_{yy} = 0,$$

it follows that

$$P_x = Q_y.$$

Second Cauchy–Riemann Equation

We have

$$P_y = u_{xy},$$

and

$$Q_x = -u_{yx}.$$

Since mixed partial derivatives commute,

$$u_{xy} = u_{yx},$$

we obtain

$$P_y = -Q_x.$$

Thus the Cauchy–Riemann equations hold.

Therefore

$$\boxed{g = u_x - iu_y}$$

is holomorphic on D .

Constructing the Holomorphic Function

Define

$$f(z) = u(z_0) + \int_{z_0}^z g(t) dt.$$

Since g is holomorphic and D is simply connected, Cauchy's theorem implies that the integral is path independent.

Therefore f is well defined.

By the fundamental theorem of complex integration,

$$\boxed{f'(z) = g(z)}.$$

Showing that $u = \Re(f)$

Write

$$f = u + iv.$$

Then

$$f' = u_x + iv_x.$$

Since

$$f' = g = u_x - iu_y,$$

we obtain

$$v_x = -u_y.$$

Comparing imaginary parts gives

$$v_y = u_x.$$

These are precisely the Cauchy–Riemann equations.

Hence u and v form a harmonic conjugate pair.

Since

$$f(z_0) = u(z_0),$$

the additive constant is chosen correctly.

Therefore

$$u = \operatorname{Re}(f).$$

Interpretation

This result establishes one of the fundamental correspondences of complex analysis:

$$\boxed{\text{Harmonic Functions} \iff \text{Real Parts of Holomorphic Functions.}}$$

The function

$$v = \operatorname{Im}(f)$$

is called a *harmonic conjugate* of u .

The holomorphic function

$$f = u + iv$$

is the analytic function associated with the harmonic function u .

Thus every harmonic function on a disc arises as the real part of a holomorphic function.

17 Problem 13: Coincidence Sets of Harmonic Functions

Statement

Suppose u, v are harmonic functions on a Riemann surface R , and let

$$E = \{z \in R : u(z) = v(z)\}.$$

Show that either

$$E = R,$$

or E has empty interior.

Give an example showing that E does not in general consist of isolated points.

Solution

Define

$$w = u - v.$$

Since both u and v are harmonic,

$$\Delta w = \Delta u - \Delta v = 0.$$

Therefore w is harmonic.

Observe that

$$E = \{z \in R : w(z) = 0\}.$$

Thus E is the zero set of the harmonic function w .

Step 1: Suppose E Contains an Open Set

Assume there exists a nonempty open subset

$$U \subset E.$$

Then

$$w = 0$$

on U .

Since harmonic functions are real analytic, the identity theorem for real analytic functions implies that a harmonic function vanishing on a nonempty open set must vanish identically on the connected component containing that set.

Hence

$$w \equiv 0.$$

Therefore

$$u = v$$

everywhere on R .

Consequently

$$E = R.$$

Step 2: Contrapositive

If

$$E \neq R,$$

then w is not identically zero.

A nonzero harmonic function cannot vanish on any nonempty open set.

Therefore

$$\text{Int}(E) = \emptyset.$$

Thus

$$\boxed{E = R \quad \text{or} \quad E \text{ has empty interior.}}$$

Example Showing that E Need Not Consist of Isolated Points

Let

$$R = \mathbb{C}.$$

Define

$$u(x + iy) = x, \quad v(x + iy) = 0.$$

Both functions are harmonic because

$$\Delta u = 0, \quad \Delta v = 0.$$

Then

$$E = \{z : u(z) = v(z)\} = \{x = 0\}.$$

Hence

$$E = i\mathbb{R}.$$

This set is an entire line.

Every point of E is an accumulation point of E .

Therefore E does not consist of isolated points.

However,

$$\text{Int}(E) = \emptyset.$$

Thus

$$\boxed{E = i\mathbb{R}}$$

provides the required example.

18 Problem 14: Four-Punctured Spheres and Cross-Ratios

Statement

Let

$$\{a_1, a_2, a_3, a_4\}$$

and

$$\{b_1, b_2, b_3, b_4\}$$

be sets of four distinct points of

$$\hat{\mathbb{C}}.$$

Show that every analytic isomorphism

$$f : \widehat{\mathbb{C}} \setminus \{a_1, a_2, a_3, a_4\} \longrightarrow \widehat{\mathbb{C}} \setminus \{b_1, b_2, b_3, b_4\}$$

extends to an analytic isomorphism

$$\widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}.$$

Using this result, find a necessary and sufficient condition for

$$\mathbb{C} \setminus \{0, 1, a\}$$

to be conformally equivalent to

$$\mathbb{C} \setminus \{0, 1, b\},$$

where

$$a, b \neq 0, 1.$$

Solution

Step 1: Extension Across the Punctures

Consider a puncture

$$a_i.$$

Since f is a biholomorphism between punctured compact Riemann surfaces, the singularity of f at a_i cannot be essential.

Indeed, by the Casorati–Weierstrass theorem (or Picard’s theorem), an essential singularity would force the image of a punctured neighborhood to be dense in the sphere, contradicting injectivity.

The singularity cannot be a pole either, because the inverse map would fail to extend holomorphically.

Therefore every puncture is removable.

Hence f extends holomorphically across all four punctures.

We obtain a holomorphic map

$$F : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}.$$

Applying the same argument to f^{-1} , we see that F^{-1} is also holomorphic.

Thus F is a biholomorphic automorphism of the Riemann sphere.

Step 2: Automorphisms of the Sphere

The automorphisms of

$$\widehat{\mathbb{C}}$$

are precisely the Möbius transformations

$$F(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0.$$

Therefore

$$f \text{ extends uniquely to a Möbius transformation.}$$

Cross-Ratios

For four points

$$(z_1, z_2, z_3, z_4)$$

their cross-ratio is

$$[z_1, z_2; z_3, z_4] = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

A fundamental theorem states that Möbius transformations preserve cross-ratios.

Thus the conformal type of a four-punctured sphere is determined by the cross-ratio of its punctures.

Application to $\mathbb{C} \setminus \{0, 1, a\}$

Write

$$\mathbb{C} \setminus \{0, 1, a\} = \widehat{\mathbb{C}} \setminus \{\infty, 0, 1, a\}.$$

The cross-ratio of the four points

$$(\infty, 0, 1, a)$$

is

$$[\infty, 0; 1, a].$$

Using the standard simplification when one point is infinity,

$$[\infty, 0; 1, a] = \frac{0 - a}{0 - 1} = a.$$

Hence the conformal type is determined by a .

However, permuting the four punctures produces the six equivalent cross-ratio values

$$a, \quad 1 - a, \quad \frac{1}{a}, \quad \frac{1}{1 - a}, \quad \frac{a - 1}{a}, \quad \frac{a}{a - 1}.$$

These are exactly the values obtained from all permutations of the four points.

Conclusion

Therefore

$$\mathbb{C} \setminus \{0, 1, a\}$$

and

$$\mathbb{C} \setminus \{0, 1, b\}$$

are conformally equivalent if and only if

$$b$$

belongs to the six-element orbit

$$\left\{ a, 1 - a, \frac{1}{a}, \frac{1}{1 - a}, \frac{a - 1}{a}, \frac{a}{a - 1} \right\}.$$

Equivalently,

$$\mathbb{C} \setminus \{0, 1, a\} \cong \mathbb{C} \setminus \{0, 1, b\}$$

if and only if a and b determine the same cross-ratio up to permutation of the punctures.

19 Theory Behind Problems 13–14: Harmonic Functions, Identity Principles, and Moduli of Four-Pointed Spheres

Problems 13 and 14 continue the development of the theory of Riemann surfaces in two complementary directions.

The first concerns the relationship between harmonic and holomorphic functions:

Harmonic Functions $\implies$ Holomorphic Functions $\implies$ Identity Principles

while the second concerns the geometry of punctured spheres:

Möbius Transformations $\implies$ Cross-Ratios $\implies$ Moduli Spaces

Together these topics form the bridge between classical complex analysis and modern algebraic geometry.

19.1 1. Harmonic Functions on Riemann Surfaces

Let

$$R$$

be a Riemann surface.

A function

$$u : R \rightarrow \mathbb{R}$$

is called *harmonic* if in every local coordinate

$$z = x + iy$$

it satisfies Laplace's equation

$$\Delta u = u_{xx} + u_{yy} = 0.$$

The operator

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

is called the Laplacian.

Physical Interpretation

Harmonic functions occur naturally in:

- Electrostatic potentials,
- Gravitational potentials,
- Steady-state heat distributions,
- Incompressible fluid flow,
- Minimal surfaces.

Thus harmonic functions arise throughout mathematics and physics.

19.2 2. Harmonic Functions are Real Analytic

One of the deepest properties of harmonic functions is:

Theorem 19.1. *Every harmonic function is real analytic.*

Consequently every harmonic function admits a convergent power series expansion

$$u(x, y) = \sum_{m, n \geq 0} a_{mn} (x - x_0)^m (y - y_0)^n.$$

This property is much stronger than smoothness.

In particular, harmonic functions satisfy an identity theorem analogous to that for holomorphic functions.

19.3 3. Identity Principle for Harmonic Functions

Suppose

$$u, v$$

are harmonic.

Define

$$w = u - v.$$

Then w is harmonic.

If

$$u = v$$

on a nonempty open set, then

$$w = 0$$

on a nonempty open set.

Since w is real analytic,

$$w \equiv 0.$$

Therefore

$$u = v$$

everywhere on the connected component.

Hence:

Two harmonic functions that agree on an open set are identical.

This is the fundamental principle used in Problem 13.

19.4 4. Difference from Holomorphic Functions

For holomorphic functions:

$$f = g$$

on a set with an accumulation point implies

$$f = g.$$

The coincidence set is either the whole domain or consists of isolated points.

For harmonic functions this is no longer true.

The coincidence set may contain entire curves.

For example,

$$u(x, y) = x, \quad v(x, y) = 0.$$

Then

$$u = v$$

precisely on

$$x = 0.$$

Thus the coincidence set is

$$i\mathbb{R},$$

the entire imaginary axis.

Hence

Coincidence sets of harmonic functions may contain curves.

This is the key distinction between Problems 11 and 13.

19.5 5. Geometry of Zero Sets of Harmonic Functions

The local structure of harmonic functions near zeros is remarkably regular.

If

$$u = \operatorname{Re}(z^n)$$

then

$$u = 0$$

consists of $2n$ rays meeting at equal angles.

Examples:

$$\operatorname{Re}(z) = x$$

has zero set

$$x = 0.$$

$$\operatorname{Re}(z^2) = x^2 - y^2$$

has zero set

$$y = \pm x.$$

$$\operatorname{Re}(z^3) = x^3 - 3xy^2$$

has six rays meeting at angles of

$$60^\circ.$$

In general, near a zero of order n ,

$$u = \operatorname{Re}(z^n) + \cdots,$$

and the nodal set consists of

$$2n$$

smooth arcs meeting equally.

Thus harmonic functions naturally generate geometric networks.

19.6 6. Harmonic Conjugates

Problem 12 established that if

$$u$$

is harmonic on a simply connected domain, then there exists a harmonic conjugate

$$v$$

such that

$$f = u + iv$$

is holomorphic.

The pair (u, v) satisfies the Cauchy–Riemann equations:

$$u_x = v_y, \quad u_y = -v_x.$$

Consequently

Harmonic Functions $\iff$ Real Parts of Holomorphic Functions.
--

This relationship lies at the heart of complex analysis.

19.7 7. The Riemann Sphere and Möbius Transformations

Problem 14 concerns the geometry of

$$\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

This space is called the *Riemann sphere*.

Its automorphism group is

$$\text{Aut}(\hat{\mathbb{C}}) = PSL(2, \mathbb{C}).$$

Every automorphism has the form

$$z \mapsto \frac{az + b}{cz + d}, \quad ad - bc \neq 0.$$

These maps are called Möbius transformations.

19.8 8. Three Points Determine a Möbius Transformation

A fundamental theorem states:

Theorem 19.2. *Given distinct points*

$$a, b, c$$

and

$$A, B, C,$$

there exists a unique Möbius transformation satisfying

$$a \mapsto A, \quad b \mapsto B, \quad c \mapsto C.$$

Therefore every triple of distinct points may be normalized to

$$0, \quad 1, \quad \infty.$$

This explains why three marked points on a sphere carry no moduli.

19.9 9. Why Four Points are Different

After fixing

$$0, \quad 1, \quad \infty,$$

a fourth point remains.

Its position cannot be removed by a Möbius transformation.

Thus a sphere with four marked points possesses one complex parameter.

This parameter is the cross-ratio.

19.10 10. Cross-Ratios

For four distinct points

$$(z_1, z_2, z_3, z_4)$$

define

$$[z_1, z_2; z_3, z_4] = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

The cross-ratio is invariant under Möbius transformations:

$$[M(z_1), M(z_2); M(z_3), M(z_4)] = [z_1, z_2; z_3, z_4].$$

Therefore it is the fundamental invariant of four-point configurations on the sphere.

19.11 11. The Moduli Space of Four Points

Every quadruple of distinct points can be transformed into

$$(\infty, 0, 1, \lambda).$$

The parameter

$$\lambda$$

determines the conformal type.

However permutations of the four points produce six equivalent values:

$$\lambda, \quad 1 - \lambda, \quad \frac{1}{\lambda}, \quad \frac{1}{1 - \lambda}, \quad \frac{\lambda - 1}{\lambda}, \quad \frac{\lambda}{\lambda - 1}.$$

Hence

$$\boxed{\mathcal{M}_{0,4} = (\mathbb{C} \setminus \{0, 1\})/S_3.}$$

This is the moduli space of four-pointed spheres.

Problem 14 computes precisely this equivalence relation.

19.12 12. Connection with Elliptic Curves

The theory becomes even richer when one considers

$$y^2 = x(x - 1)(x - \lambda).$$

The branch points are

$$0, \quad 1, \quad \lambda, \quad \infty.$$

The associated Riemann surface is a torus.

Thus the parameter

$$\lambda$$

controls the complex structure of an elliptic curve.

This family is called the *Legendre family*.

Consequently

$$\boxed{\text{Cross-Ratios} \implies \text{Elliptic Curves.}}$$

19.13 13. Modular Functions

Every complex torus can be written as

$$T_\tau = \mathbb{C}/\langle 1, \tau \rangle.$$

The Legendre parameter and the torus parameter are related by

$$\lambda = \lambda(\tau),$$

where $\lambda(\tau)$ is the modular lambda function.

Hence

$$\boxed{\text{Four-Pointed Spheres} \iff \text{Complex Tori.}}$$

This connection is one of the foundations of modern algebraic geometry.

19.14 Grand Synthesis

The conceptual flow behind Problems 13–14 is

$$\boxed{\text{Harmonic Functions} \implies \text{Holomorphic Functions} \implies \text{Identity Principles}}$$

and simultaneously

$$\boxed{\text{Möbius Transformations} \implies \text{Cross-Ratios} \implies \text{Moduli Spaces}}$$

which ultimately leads to

$$\boxed{\text{Cross-Ratios} \implies \text{Elliptic Curves} \implies \text{Complex Tori} \implies \text{Modular Forms.}}$$

Thus Problems 13–14 mark the transition from classical complex analysis to the modern theory of moduli spaces, elliptic curves, and algebraic geometry.